

- 1 Guinea pigs, *Cavia porcellus*, vary in the length and colour of their fur.

Fig. 1.1 shows a guinea pig with short black fur.



Fig. 1.1

Two genes that determine the length and colour of the fur occur at the **A/a** locus and the **B/b** locus. These two gene loci are on separate autosomal chromosomes.

- The allele **A** results in short fur.
- The allele **a** results in long fur.
- **A** is dominant to **a**.
- The allele **B** results in black fur.
- The allele **b** results in chocolate fur.
- **B** is dominant to **b**.

- (a) (i) List **all** the possible genotypes of a guinea pig with short black fur.

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.....
..... [2]

- (ii) A test cross could be used to determine the genotype of a female guinea pig with short black fur.

Describe the **phenotype** of the male guinea pig that could be used to carry out this test cross.

..... [1]



- (b) A black guinea pig with long fur that was homozygous at both loci was crossed with a chocolate guinea pig with short fur that was homozygous at both loci. The F₁ offspring of this cross had short black fur. F₁ offspring were mated together to produce the F₂ offspring.

Complete the Punnett square to:

- show the cross between the F₁ offspring
- predict the F₂ offspring genotypes.

You should include the gametes in your answer.

State the ratio of F₂ offspring phenotypes. You should include a key to link phenotypes to genotypes.

ratio of F₂ offspring phenotypes:

[4]

- (c) Some genes in guinea pigs are structural genes and some are regulatory genes.

Describe the difference between a structural gene and a regulatory gene.

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..... [2]

[Total: 9]